

海洋航行器六自由度非线性建模 (Fossen 1995 中英双语对照版)

海洋航行器六自由度非线性建模 (Nonlinear Modelling of Marine Vehicles in 6 Degrees of Freedom)

中英双语对照精读版 (Bilingual Parallel Edition)

英文原题 (Original Title): Nonlinear Modelling of Marine Vehicles in 6 Degrees of Freedom

论文作者 (Authors): Thor I. Fossen, Ola-Erik Fjellstad

作者单位 (Affiliation): 挪威理工学院工程控制系 (The Norwegian Institute of Technology, Department of Engineering Cybernetics, NTNU)

发表刊物 (Publication): *Journal of Mathematical Modelling of Systems*, Vol. 1, No. 1, 1995, pp. 17 - 27

学术定位: 海洋航行器 (表面船舶、潜艇、ROV/AUV) 六自由度 (6-DOF) 非线性动力学统一机理建模奠基作。

摘要 (Abstract)

[EN] In this paper a unified framework for vectorial parameterisation of inertia, Coriolis and centrifugal, and hydrodynamic added mass forces for marine vehicles in 6 degrees of freedom (DOF) is presented. Emphases is placed on representing the 6 DOF nonlinear marine vehicle equations of motion in vector form satisfying certain matrix properties like symmetry, skew-symmetry and positive definiteness. For marine vehicles, this problem was first addressed by Fossen [1]. The proposed representations in this paper are based on extensions of this work. The main results are presented as two theorems. The first theorem is derived by applying Newton's equations whereas the second theorem reviews the main results of Sagatun and Fossen [6] where the Lagrangian formalism is applied.

[中文] 本文提出了一个用于海洋航行器在六自由度 (6-DOF) 运动空间中对刚体惯性力、科氏力与向心力以及流体动力附加质量力进行矢量参数化描述的统一理论框架。研究重点在于将六自由度非线性海洋航行器运动方程表达为紧凑的矢量形式, 并证明该形式满足对称性 (Symmetry)、反对称性 (Skew-symmetry) 以及正定性 (Positive definiteness) 等重要的系统矩阵代数性质。在海洋工程与控制领域, 该问题最早由 Fossen [1] 展开探索, 本文提出的矢量化表达是基于前期工作的深入拓展。本文的核心理论贡献凝练为两个基础定理: 定理 1 严格基于牛顿-欧拉方程 (Newtonian approach) 推导得出; 定理 2 则基于拉格朗日分析力学形式 (Lagrangian formalism) 对流体动能及系统能量守恒特性进行回顾与统一。

关键词 (Key words): 建模 (Modelling), 海洋航行器 (Marine vehicles), 运动方程 (Equations of motion), 拉格朗日力学 (Lagrangian mechanics), 牛顿力学 (Newtonian mechanics).

1. 引言 (1. Introduction)

[EN] In order to describe the motion of a marine vehicle in 6 degrees of freedom (DOF), 6 independent coordinates representing position and attitude are necessary. The 6 different motion components are defined as: surge, sway, heave, roll, pitch and yaw (see Table 1).

[中文] 为了在三维水下空间中完整描述海洋航行器的六自由度（6-DOF）空间运动，必须引入 6 个独立的广义坐标来表示航行器的空间位置与姿态角。这 6 个运动分量分别定义为：纵荡（Surge）、横荡（Sway）、垂荡（Heave）、横滚（Roll）、俯仰（Pitch）与偏航（Yaw）（见表 1）。

表 1：海洋航行器六自由度运动分量定义 (Table 1: 6 DOF motion components)

自由度 (DOF)	运动分量与物理方向 (Motions)	作用力与力矩 (Forces & Moments)	附体系速度 (Body-Fixed Velocities)	惯性系位置与姿态 (Inertial Pos & Angles)
1	motions in the x-direction (纵荡, surge)	X	u	x
2	motions in the y-direction (横荡, sway)	Y	v	y
3	motions in the z-direction (垂荡, heave)	Z	w	z
4	rotation about the x-axis (横滚, roll)	K	p	ϕ
5	rotation about the y-axis (俯仰, pitch)	M	q	θ
6	rotation about the z-axis (偏航, yaw)	N	r	ψ

[EN] Hence the general motion of a marine vehicle in 6 DOF can be described by the following vectors:

[中文] 据此，海洋航行器在六自由度空间中的广义运动状态可以用以下标准矢量表达：

$$\eta = \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix}, \quad \eta_1 = [x, y, z]^T, \quad \eta_2 = [\phi, \theta, \psi]^T$$
$$\nu = \begin{bmatrix} \nu_1 \\ \nu_2 \end{bmatrix}, \quad \nu_1 = [u, v, w]^T, \quad \nu_2 = [p, q, r]^T$$
$$\tau = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}, \quad \tau_1 = [X, Y, Z]^T, \quad \tau_2 = [K, M, N]^T$$

[EN] where η denotes the earth-fixed position and attitude vector, ν denotes the body-fixed linear and angular velocity vector and τ is used to describe the forces and moments acting on the vehicle in the body-fixed frame. The body-fixed and inertial reference frames are shown in Figure 1.

[中文] 其中 η 表示地球固连惯性坐标系（北-东-地 NED 坐标系）下的位置与欧拉角姿态矢量； ν 表示固连于航行器机体的附体坐标系（前-右-下 FRD 坐标系）下的线速度与角速度矢量； τ 表示作用在航行器机体上的广义外部控制力与控制力矩螺旋（Wrench）。附体坐标系 $\{b\}$ 与惯性参考系 $\{n\}$ 的几何关系如图 1 所示。

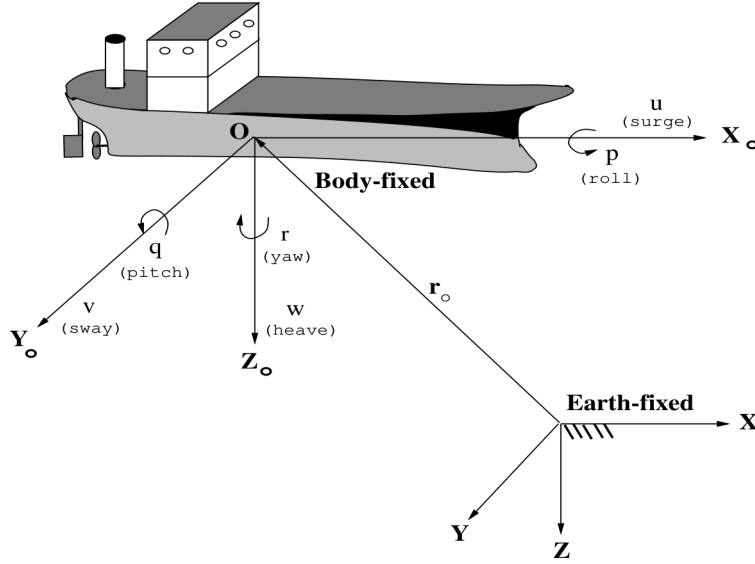


图 1: 海洋航行器附体坐标系与惯性参考系几何关系 (Figure 1: Body-fixed and earth-fixed reference frames)

运动学变换关系 (Kinematics)

$$\dot{\eta}_1 = J_1(\eta_2)\nu_1, \quad \dot{\eta}_2 = J_2(\eta_2)\nu_2$$

$$\dot{\eta} = J(\eta)\nu = \begin{bmatrix} J_1(\eta_2) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & J_2(\eta_2) \end{bmatrix} \begin{bmatrix} \nu_1 \\ \nu_2 \end{bmatrix}$$

2. 牛顿力学方法与刚体动力学 (2. Newtonian Approach)

[EN] The motion of a rigid body with respect to a body-fixed rotating reference frame $X_0Y_0Z_0$ with origin O is given by Newton's laws:

[中文] 刚体相对于原点为 O 的附体旋转坐标系 $X_0Y_0Z_0$ 的空间运动由牛顿-欧拉方程严格给出:

$$m[\dot{\nu}_1 + \nu_2 \times \nu_1 + \dot{\nu}_2 \times r_G + \nu_2 \times (\nu_2 \times r_G)] = \tau_1$$

$$I_0\dot{\nu}_2 + \nu_2 \times (I_0\nu_2) + m r_G \times (\dot{\nu}_1 + \nu_2 \times \nu_1) = \tau_2$$

[EN] Defining the skew-symmetric cross-product operator $S(\lambda)$ such that $S(\lambda)a = \lambda \times a$, the 6 DOF rigid-body equations can be expressed in vector form as:

[中文] 定义满足 $S(\boldsymbol{\lambda})\mathbf{a} = \boldsymbol{\lambda} \times \mathbf{a}$ 的三维叉乘反对称算子 $S(\boldsymbol{\lambda})$ ，六自由度刚体方程可表达为紧凑的矢量形式：

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

其中刚体质量惯性矩阵 $\mathbf{M}_{RB} \in \mathbb{R}^{6 \times 6}$ 具有对称正定性：

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_{3 \times 3} & -mS(\mathbf{r}_G) \\ mS(\mathbf{r}_G) & \mathbf{I}_0 \end{bmatrix} = \mathbf{M}_{RB}^T > 0$$

刚体科氏力与向心力矩阵 $\mathbf{C}_{RB}(\boldsymbol{\nu})$ 能够参数化为严格反对称矩阵：

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -mS(\boldsymbol{\nu}_1) - mS(\boldsymbol{\nu}_2)S(\mathbf{r}_G) \\ -mS(\boldsymbol{\nu}_1) + mS(\mathbf{r}_G)S(\boldsymbol{\nu}_2) & -S(\mathbf{I}_0\boldsymbol{\nu}_2) \end{bmatrix}$$

满足关键性质：

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = -\mathbf{C}_{RB}^T(\boldsymbol{\nu}), \quad \boldsymbol{\nu}^T \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = 0, \quad \forall \boldsymbol{\nu} \in \mathbb{R}^6$$

3. 六自由度海洋航行器水动力统一建模 (3. Marine Vehicles in 6 DOF)

[EN] The general 6 DOF nonlinear marine vehicle equations of motion are given by:

[中文] 六自由度海洋航行器非线性动力学统一标准方程表达为：

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau}$$

3.1 质量矩阵与附加质量 (Total Mass & Added Mass)

$$\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A \in \mathbb{R}^{6 \times 6}, \quad \mathbf{M} = \mathbf{M}^T > 0$$

流体附加质量矩阵 $\mathbf{M}_A$ 具有对称正定性：

$$\mathbf{M}_A = - \begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix} = \mathbf{M}_A^T > 0$$

3.2 水动力科氏向心矩阵与能量守恒定理 (Theorem 1)

将附加质量矩阵分块写为 $\mathbf{M}_A = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}$ 。

[Theorem 1] The hydrodynamic Coriolis and centripetal matrix $\mathbf{C}_A(\boldsymbol{\nu})$ can always be parameterised as a skew-symmetric matrix:

[定理 1] 水动力附加质量诱导的科氏与向心力矩阵可严格参数化为反对称矩阵形式：

$$\mathbf{C}_A(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -S(\mathbf{A}_{11}\boldsymbol{\nu}_1 + \mathbf{A}_{12}\boldsymbol{\nu}_2) \\ -S(\mathbf{A}_{11}\boldsymbol{\nu}_1 + \mathbf{A}_{12}\boldsymbol{\nu}_2) & -S(\mathbf{A}_{21}\boldsymbol{\nu}_1 + \mathbf{A}_{22}\boldsymbol{\nu}_2) \end{bmatrix}$$

总科氏向心矩阵 $C(\boldsymbol{\nu}) = C_{RB}(\boldsymbol{\nu}) + C_A(\boldsymbol{\nu})$ 恒满足：

$$C(\boldsymbol{\nu}) = -C^T(\boldsymbol{\nu}), \quad \boldsymbol{\nu}^T C(\boldsymbol{\nu}) \boldsymbol{\nu} = 0, \quad \forall \boldsymbol{\nu} \in \mathbb{R}^6$$

$$\mathbf{s}^T [\dot{\mathbf{M}} - 2C(\boldsymbol{\nu})] \mathbf{s} = 0, \quad \forall \mathbf{s} \in \mathbb{R}^6$$

该性质证明了系统无源性，构成了现代水下非线性运动控制（反步法、李雅普诺夫稳定性设计）的数学基石。

3.3 流体阻尼与重浮力恢复力 (Damping & Restoring)

$$D(\boldsymbol{\nu}) = D_{lin} + D_{quad}(\boldsymbol{\nu}) > 0, \quad \forall \boldsymbol{\nu} \neq \mathbf{0}$$

流体机械能耗散性保证了系统的固有阻尼稳定性。静水重浮力恢复力 $\mathbf{g}(\boldsymbol{\eta})$ 由稳心高及重心浮心空间偏置决定，为横滚与俯仰提供天然回复刚度。

4. 结论 (4. Conclusions)

[EN] A unified framework for vectorial parameterisation of the 6 DOF equations of motion of marine vehicles has been presented. The matrix properties including symmetry of the inertia matrix, skew-symmetry of the Coriolis and centripetal matrix, and positive definiteness of the damping matrix are explicitly derived and proven.

[中文] 本文系统建立了海洋航行器六自由度非线性运动方程的统一矢量化参数化框架。严格推导并证明了惯性矩阵的对称性、科氏向心矩阵的反对称性以及流体阻尼矩阵的正定耗散性，为海洋机器人现代模型控制提供了标准的理论范式。